

研究方向

我是北京大学软件与微电子学院硕士研究生，导师为马连韬教授和王亚沙教授。我的研究方向为医疗健康人工智能，重点关注医疗大语言模型及多智能体协作应用、时序电子病历数据建模，以及面向临床数据分析的可复用基准与工具基础设施。

- (1) 医疗大语言模型：构建并评估面向临床预测、推理、检索和决策支持的大语言模型驱动的多智能体系统。
- (2) 时序电子病历数据建模：面向纵向、稀疏和多模态 EHR 数据进行建模，关注临床可靠性与复用。
- (3) 基准、工具包与协议：构建开放资源，提高医疗健康 AI 评估的可复现性与实际可用性。

教育经历

北京大学

软件与微电子学院·软件工程硕士

2024.09 - 至今

关注医疗健康 AI、医疗大语言模型与时序电子病历数据建模研究，导师为马连韬教授和王亚沙教授。

北京航空航天大学

软件学院·软件工程学士

2019.09 - 2024.06

GPA: 3.83 (排名: 9/187, 前 5%)。

论文发表 (* 表示共同一作; † 表示通讯作者)

Core Work

[WWW 2025

- CCF A]

ColaCare: Enhancing Electronic Health Record Modeling through Large Language Model-Driven Multi-Agent Collaboration

Zixiang Wang*, Yinghao Zhu*, Huiya Zhao*, Xiaochen Zheng, Tianlong Wang, Wen Tang, Yasha Wang†, Chengwei Pan, Ewen Harrison, Junyi Gao†, Liantao Ma†

ACM International World Wide Web Conference (WWW), 2025

[CHI 2026

- CCF A]

Augmenting Clinical Decision-Making with an Interactive and Interpretable AI Copilot: A Real-World User Study with Clinicians in Nephrology and Obstetrics

Yinghao Zhu*, Dehao Sui*, Zixiang Wang*, Xuning Hu, Lei Gu, Yifan Qi, Tianchen Wu, Ling Wang, Yuan Wei, Wen Tang, Zhihan Cui, Yasha Wang, Lequan Yu, Ewen Harrison, Junyi Gao, Liantao Ma†

ACM CHI Conference on Human Factors in Computing Systems (CHI), 2026

[KDD 2024

- CCF A]

RetCare: Towards Interpretable Clinical Decision Making through LLM-Driven Medical Knowledge Retrieval

Zixiang Wang*, Yinghao Zhu*, Junyi Gao, Xiaochen Zheng, Yuhui Zeng, Yifan He, Bowen Jiang, Wen Tang, Ewen Harrison, Chengwei Pan, Liantao Ma†, Ling Wang†

Artificial Intelligence and Data Science for Healthcare: Bridging Data-Centric AI and People-Centric Healthcare (KDD 2024 AIDSH Workshop), 2024

[npj Digital

Medicine 2026]

ClinicRealm: Re-evaluating Large Language Models with Conventional Machine Learning for Non-Generative Clinical Prediction Tasks

Yinghao Zhu*, Junyi Gao*, Zixiang Wang*, Weibin Liao*, Xiaochen Zheng, Lifang Liang, Miguel O. Bernabeu, Yasha Wang, Lequan Yu, Chengwei Pan†, Ewen Harrison†, Liantao Ma†

npj Digital Medicine, 2026

[npj Digital

Medicine 2026 -

Minor Revision]

HealthFlow: Automating electronic health record analysis via a strategically self-evolving multi-agent framework

Yinghao Zhu*, Zixiang Wang*, Yifan Qi*, Lei Gu, Dehao Sui, Haoran Hu, Xichen Zhang, Ziyi He, Junjun He, Liantao Ma†, Lequan Yu†

Preprint, 2025

[STAR Protocols

2025]

Protocol for Processing Multivariate Time-series Electronic Health Records of COVID-19 Patients

Zixiang Wang*, Yinghao Zhu*, Dehao Sui, Tianlong Wang, Yuntao Zhang, Yasha Wang, Chengwei Pan, Junyi Gao†, Liantao Ma†, Ling Wang†, Xiaoyun Zhang†

STAR Protocols, 2025

LLM Agents for Healthcare

[KDD 2024]

EHRFlow: A Large Language Model-Driven Iterative Multi-Agent Electronic Health Record Data Analysis Workflow

Co-authors

Artificial Intelligence and Data Science for Healthcare: Bridging Data-Centric AI and People-Centric Healthcare (KDD 2024 AIDSH Workshop), Oral, 2024

[CIKM 2024]

EMERGE: Enhancing Multimodal Electronic Health Records Predictive Modeling with Retrieval-Augmented Generation

Co-authors

ACM International Conference on Information and Knowledge Management (CIKM), 2024

Toolkits & Platforms & Benchmarks

[KDD 2026]	OneEHR: Reproducible and AI Agent-Ready Longitudinal EHR Analysis Toolkit Co-authors <i>ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD-26) Hands-On Tutorials</i> , 2026
[SAIL 2025]	AIcCare: An AI-Clinician Interaction System for Transparent and Actionable Clinical Decision Support Co-authors <i>The Symposium on Artificial Intelligence in Learning Health Systems (SAIL)</i> , Best abstract nominee, 2025
[NeurIPS 2025]	MedAgentBoard: Benchmarking Multi-Agent Collaboration with Conventional Methods for Diverse Medical Tasks Co-authors <i>Conference on Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track</i> , 2025
[Cell Patterns 2024]	A Comprehensive Benchmark For COVID-19 Predictive Modeling Using Electronic Health Records in Intensive Care Co-authors <i>Cell Patterns</i> , 2024
[STAR Protocols 2024]	Protocol to process follow-up electronic medical records of peritoneal dialysis patients to train AI models Co-authors <i>STAR Protocols</i> , 2024
[Preprint 2026]	Auditing medical multi-agent AI reveals risks of false consensus Co-authors <i>Preprint</i> , 2026

Healthcare Modeling

[KDD 2025]	Learnable Prompt as Pseudo-Imputation: Rethinking the Necessity of Traditional EHR Data Imputation in Downstream Clinical Prediction Co-authors <i>ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)</i> , 2025
[CIKM 2024]	PRISM: Mitigating EHR Data Sparsity via Learning from Missing Feature Calibrated Prototype Patient Representations Co-authors <i>ACM International Conference on Information and Knowledge Management (CIKM)</i> , 2024

代表项目

▷ 小雅医生 (AIcCare): 面向电子健康记录分析的 AI 临床决策支持平台, 部署于北京大学 Xplore “临床医学 + X”平台	2025
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荣誉与奖励

▷ 北京航空航天大学优秀毕业生, 北京航空航天大学	2024
▷ 国家励志奖学金, 北京航空航天大学	2022, 2023

学术服务

▷ 审稿人, AAAI 2026, 2027	2025, 2026
▷ 审稿人, npj Digital Medicine	2026
▷ 审稿人, npj Health Systems	2026
▷ 审稿人, Scientific Reports	2026